Emergent Network Dynamics and the 9.15% Redshift Correction: A Resolutive Framework for High-Redshift Spectroscopic Anomalies

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Subject: Formalism for Emergent Spacetime via Discrete Informational Topology

I. Abstract

The contemporary field of cosmology is currently navigating a period of profound empirical dissonance. The traditional \LambdaCDM paradigm faces two potentially fatal challenges: the "Early Universe Crisis" of mature z > 14 galaxies and the "Hubble Tension." We formalize the **Informational Connectivity Metric (ICM)**, a framework where the spacetime manifold is treated as the continuum limit of a self-organizing discrete substrate with a finite capacity N \approx 10^{140}. By defining physical distance as an inverse function of informational coupling, we derive universal constants as emergent network properties and identify a 9.11% propagation lag that resolves the expansion discrepancy as a native structural property of the network.

II. The Spectroscopic Challenge of the High-Redshift Universe

The operation of the James Webb Space Telescope (JWST) has provided spectroscopic data that contradicts the core assumptions of hierarchical galactic assembly. In the standard \LambdaCDM model, structure formation is a slow, bottom-up process. However, the detection of massive, mature systems at z > 10 suggests a fundamental deficiency in our understanding of early-time physics.

1. Spectroscopic Confirmation of "Impossible" Galaxies

The presence of **JADES-GS-z14-0** at z = 14.32 represents a critical breaking point. Existing only ~290 million years after the Big Bang, it spans over 1,600 light-years and exhibits stellar masses previously thought impossible for this epoch.

Galaxy Candidate	Redshift (z)	Stellar Mass (M_sun)	Cosmic Age (Myr)
JADES-GS-z14-0	14.32	10^8.2	~290
GLASS-z12	12.47	10^9	~350
CEERS2-588	11.04	10^9	~400

III. The Mathematical Framework (ICM)

To resolve these anomalies, we move toward a "substrate-first" ontology where fundamental constants are derived properties.

1. Scaling Law and Inverse Weight Rule

We link small-scale geometry to universal boundary conditions through the substrate capacity $N \approx 10^{140}$.

The physical distance is defined as: **$d_{ij} = \frac{\ell_p}{\bar{W}_{ij} + \epsilon_i}$**

2. Derivation of Emergent Constants

Fundamental constants are interpreted as properties of the network's update frequency and stiffness.

- **Speed of Light (c):** $c = \frac{\Delta x_{min}}{\Delta \tau}$
- **Gravitational Constant (G):** $G = \frac{\Delta \tau}{\gamma N}$

IV. Resolutive Mechanisms

The ICM provides a dual-resolution for current cosmological crises:

1. **Topological Metric Cavitation:** Explains "instant" galaxy formation via supercritical Hopf bifurcations in the early metric.
2. **The 9.15% Hubble Solution:** Recovers the Hubble tension as **Temporal Inertia**—the propagation latency inherent in translating discrete updates into a smooth manifold.

V. Falsifiable "Smoking Gun" Predictions

The ICM predicts a **strictly positive sign** for cosmological redshift drift at $z > 2$, directly contradicting the negative drift predicted by ΛCDM. This test is achievable via the Extremely Large Telescope (ELT).

VI. Conclusion

As $N \to 10^{140}$, the discrete geometry converges toward the continuous metric of General Relativity. The ICM offers a finite, renormalizable bridge across dimensions.

VII. Appendix: Formal Mathematical Formalisms

The following sections are reserved for the high-resolution formalisms of the Informational Connectivity Metric.

A. Substrate Capacity and Scaling Laws

1. emergent planck length:

2. emergent speed of light:

3. emergent gravitational constant:

4. temporal inertia / Hubble-lag:

$$\ell_p^{emergent} = \left(\frac{V}{N}\right)^{1/3},$$

$$c_{emergent} = \frac{\ell_p^{emergent}}{\Delta\tau}$$

$$G_{emergent} = \frac{\Delta\tau}{\gamma N},$$

$$\frac{\Delta t_i}{dt} \sim \frac{1}{N},$$

B. The Unified Informational Metric

1. Substrate Capacity:

2. Emergent Planck Resolution:

3. Emergent Light Speed (Causal Update):

4. Adjacency Distance with Zero-Point Buffer:

5. Network Elastic Energy:

6. Background-Independent Curvature (Ollivier–Ricci):

7. Graph Laplacian (Discrete Geometry Operator):

8. Ensemble Emergent Metric:

9. Emergent Einstein Tensor:

10. Informational Stress–Energy Density:

11. Emergent Gravitational Constant (Network Stiffness):

12. Discrete Einstein Equation:

13. Master Informational Action:

14. Gauge Symmetry (Emergent Forces):

15. Gauge Curvature Tensor:

16. Topological Particle Charge:

17. Quantum Evolution on the Network:

18. Hubble Latency (Spectral Prediction):

19. Spectral Gap Scaling:

20. Continuum Limit (Recovery of GR):

21. Node Position Update:

22. Node Velocity Update:

$$N \approx 10^{140}, \qquad V = \text{constant substrate volume}$$

$$\ell_p = \left(\frac{V}{N}\right)^{1/3}$$

$$c = \frac{\ell_p}{\Delta\tau}$$

$$d_{ij} = \frac{\ell_p}{W_{ij} + \epsilon_i}$$

$$E_{ij} = \frac{1}{2}k_s(d_{ij} - \ell_p)^2$$

$$\kappa_{ij} = 1 - \frac{W_1(m_i, m_j)}{d_{ij}}$$

$$L_{ij} = D_{ij} - W_{ij}$$

$$g_{ij}^{edge} = \left\langle \frac{\sum_\mu W_{ij}^\mu (x_i^\mu - x_j^\mu)^2}{d_{ij}^2} \right\rangle_{\text{ensemble}}$$

$$G_i = \frac{1}{|N_i|}\sum_{j \in N_i}\left(\kappa_{ij} - \frac{1}{2}\bar{\kappa}_i g_{ij}^{edge}\right)$$

$$T_i = m_i + \sum_{k \in H}\widetilde{W}_{ik}\epsilon_k(t) + \sum_\alpha \langle \psi_i | H_i^{gauge,\alpha} + H_{ijk}^{gauge} | \psi_i \rangle$$

$$\nabla_i T_i = 0$$

$$G = \frac{\ell_p^2}{k_s \Delta\tau}$$

$$G_i = \frac{8\pi G}{c^4} T_i$$

$$S = \sum_{ij} k_s(d_{ij} - \ell_p)^2 + \sum_{ij}\kappa_{ij} + \sum_i T_i$$

$$\delta S = 0$$

$$W_{ij} \to e^{i\theta_{ij}} W_{ij}$$

$$F_{ij} = \partial_i A_j - \partial_j A_i$$

$$Q_i = \sum_{\triangle ijk} \epsilon_{ijk} W_{ij} W_{jk} W_{ki}$$

$$i\eta \frac{d\psi_i}{d\tau} = \sum_j W_{ij} \psi_j$$

$$\frac{\Delta H}{H} = \frac{\lambda_2}{\langle k \rangle}$$

$$\lambda_2 \approx \frac{\log N}{N^{1/3}}$$

D. Node Evolution and Continuum Closure

$$\lim_{N \to 10^{140}} g_{ij}^{edge} \to g_{\mu\nu}(x), \quad \lim_{N \to 10^{140}} G_{ij} \to G_{\mu\nu}(x), \quad \lim_{\ell_p \to 0} \frac{\kappa_{ij}}{\ell_p^2} \to R_{\mu\nu}$$

$$x_i(t + \Delta t_i) = x_i(t) + v_i(t)\Delta t_i + \frac{1}{2} a_i(t) \Delta t_i^2$$

$$v_i(t + \Delta t_i) = v_i(t) + a_i(t)\Delta t_i$$

References

[1] Flint, M. H., *Project-ICM-9.1*, GitHub: https://github.com/flintmitch7/Project-ICM-9.1 [2] Loeffen, R., *Cosmic Influx Theory and the 9.15% Correction* (2025). [3] JADES Collaboration, *Spectroscopic confirmation of galaxies at z > 14* (2024).